IBM Db2 TechX Summit 2026 — Architecture Specification

Agentic AI-Powered Enterprise Database Security, Anomaly Detection & Incident Response Platform

Project Slug

Direct PR Link

Frameworks

Db2 Connector

1. Executive Summary & Problem Landscape

Enterprise relational databases represent the ultimate target for cyber adversaries, storing mission-critical financial ledgers, customer PII, and administrative credentials. Traditional SOC monitoring creates severe operational liabilities: database audit facilities generate tens of thousands of raw logs daily, overwhelming security analysts with false positives, while "low-and-slow" insiders evade static perimeter alarms. Manually correlating cross-session queries and database telemetry typically requires 4 to 6 hours per incident.

> **Core Architectural Differentiator:** Db2Sentinel AI elevates **IBM Db2** from a passive backend repository into the active security intelligence layer. IBM Db2 serves as the single source of truth for normalized event streams, deterministic anomaly rules, multi-agent investigation records, and human-in-the-loop audit trails.

2. End-to-End System Architecture

The platform is engineered across eight distinct functional layers, ensuring strict separation of concerns, enterprise scalability, and defense-in-depth:



1. IBM Db2 Kernel

Native db2audit facility & Activity Event Monitors capturing SQL DDL/DML execution in real-time.

2. Detection & Risk Engine

6 deterministic security rules coupled with a 4-factor mathematical scoring formula (0–100).

3. LangGraph Agents

6 stateful AI agents orchestrating detection, timeline correlation, risk analysis, and report generation.

3. IBM Db2 Integration & Data Architecture

All database communication is conducted using the official IBM C-extension driver `ibm_db`. Telemetry ingestion is segregated across dual paths:

- **Path A — db2audit Facility:** Ingests kernel-level authentication failures (`SQL30082N`), privilege violations (`CHECKING`), and administrative context changes.
- **Path B — Activity Event Monitors (`WRITE TO TABLE`):** Ingests granular SQL query syntax, compilation cost metrics, rows read/returned, and application handles.

Enterprise Relational Schema (6 Core Entities)

Db2 Table	Primary Keys / Indexes	Functional Purpose in SOC Architecture
SENTINEL_EVENTS	`event_id` (B-Tree on timestamp, user, IP)	Normalized telemetry store capturing all connection, query, and administrative events.
SENTINEL_ALERTS	`alert_id` (Foreign Key: `event_id`)	Rule-triggered alarms classifying security threshold breaches and initial severity.

SENTINEL_INCIDENTS	`incident_id` (Status: OPEN, ESCALATED, CLOSED)	Correlated security cases grouping multi-event attack timelines and blast radius.
SENTINEL_INVESTIGATIONS	`investigation_id` (FK: `incident_id`)	Agent-specific findings, attack hypotheses, and MITRE ATT&CK technique citations.
SENTINEL_APPROVALS	`approval_id` (FK: `incident_id`)	Human-in-the-loop staging table storing requested containment commands and DBA reviews.
SENTINEL_AUDIT_LOG	`audit_id` (Timestamp, SHA-256 hash)	Immutable cryptographic audit trail recording all reviewer approvals and actions.

4. Cybersecurity Detection & 4-Factor Risk Engine

Rather than relying on opaque LLM predictions for initial detection, Db2Sentinel AI deploys a hybrid deterministic detection engine combining 6 rules with statistical variance calculations:

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-- Transparent 4-Factor Mathematical Risk Formula (0 to 100): Risk Score = (0.30 * Behavioral_Anomaly) + (0.25 * Privileged_Execution) + (0.25 * Volume_Frequency) + (0.20 * Auth_Failures) Severity Buckets: * 00 – 24 : LOW (Informational variance; automated logging) * 25 – 49 : MEDIUM (Isolated anomaly or minor off-hours query) * 50 – 74 : HIGH (Multiple failed logins or sensitive table exploration) * 75 – 100: CRITICAL (Active exfiltration or compromised administrative account)
```

5. LangGraph Multi-Agent Architecture & Governance

The investigation workflow is orchestrated as a compiled LangGraph `StateGraph`. Each agent node executes specialized reasoning with strict JSON schema boundaries:

Agent Node	Framework	Operational Responsibility
Detection Agent	LangGraph Node	Parses rule alerts, extracts IoCs, and establishes incident severity.
Evidence Agent	LangGraph Node	Queries Db2 for historical connection baselines using immutable `event_id` citations.
Threat Intel Agent	LangGraph Node	Enriches findings against MITRE ATT&CK techniques (T1110, T1078, T1005, T1505).

Risk Agent	LangGraph Node	Evaluates the 4-factor formula and produces transparent score rationales.
Response Agent	LangGraph Node	Generates targeted Db2 remediation scripts (`FORCE APPLICATION`, `REVOKE`).
Approval Checkpoint	LangGraph Interrupt	**Human-in-the-Loop Barrier:** Freezes workflow until certified DBA signs off.
Report Agent	LangGraph Node	Synthesizes executive incident reports and records cryptographic audit logs.

6. Security Model & Defense-in-Depth

- **Least-Privilege Database Access:** The ingestion service holds `INSERT` privileges only on `SENTINEL_EVENTS`. The agent engine operates with read-only `SELECT` permissions. Administrative remediation requires signed human approval tokens.
- **Prompt Injection Defense:** All SQL statements, table identifiers, and user parameters are strictly encapsulated as structured JSON data values, never interpolated into raw LLM prompt strings.
- **Deterministic Fallback:** If an LLM API is unavailable, the deterministic rule engine independently computes scores and containment scripts, ensuring air-gapped high availability.

7. Hackathon Submission & Verification Summary

Component	Technology	Verification Status
Official PR	GitHub Pull Request	PR #81 (Submitted & Validated)
Submission Validator	Python 3.11	Passed all 5 official repository checks
Automated Tests	pytest	10/10 passed in 4.26s (CRUD, Rules, Risk, Workflow)
Frontend Build	Vite + TypeScript	2,173 modules compiled cleanly with 0 errors
Database Integration	ibm_db + Db2 11.5+	Dual-mode verified (Live Db2 + local relational engine)

Db2Sentinel AI — IBM Db2 TechX Summit 2026 Hackathon (Round 2)

Confidential & Technical Architecture Specification